1. AIM

To analyze an E-Commerce Sales dataset and create three visualizations—bar chart, line chart, and pie chart—using Python, Power BI, and Tableau to understand product sales, analyze monthly revenue trends, and identify the contribution of different product categories to total sales.

2. PROBLEM STATEMENT

E-Commerce businesses generate large amounts of data related to products, sales, quantities, prices, and order dates. Analyzing this data manually is difficult. This project uses Python libraries such as Pandas and Matplotlib, along with Power BI and Tableau, to process sales data and create visualizations. The visualizations help identify high-revenue products, analyze monthly sales trends, and understand category-wise revenue contribution.

3. SOFTWARE REQUIREMENTS

• Platform: Google Colab, Power BI Desktop, Tableau Desktop

• Programming Language: Python

• Dataset Format: CSV (.csv)

• Libraries: Pandas, Matplotlib

• Input Dataset: ecommerce_sales.csv

4. DATASET DESCRIPTION

The sample dataset contains 20 order records. Sales is calculated by multiplying Quantity by Unit_Price.

Column Name	Description
Order_ID	Unique order identifier
Order_Date	Date when the order was placed
Product	Product name
Category	Product category
Quantity	Number of units sold
Unit_Price	Price of one unit in rupees
Sales	Quantity × Unit_Price

5. SAMPLE DATASET

Create a file named ecommerce_sales.csv and enter the following data. Use this same file in Python, Power BI, and Tableau.

```
Order_ID,Order_Date,Product,Category,Quantity,Unit_Price
101,2026-01-05,Laptop,Electronics,2,50000
102,2026-01-10,Mobile,Electronics,3,20000
103,2026-01-15,Chair,Furniture,4,3000
104,2026-01-20,Table,Furniture,2,7000
105,2026-02-03,Headphones,Electronics,5,2000
106,2026-02-08,Keyboard,Accessories,4,1000
107,2026-02-12,Mouse,Accessories,6,500
108,2026-02-18,Mobile,Electronics,2,20000
109,2026-03-04,Laptop,Electronics,1,50000
110,2026-03-09,Chair,Furniture,3,3000
111,2026-03-14,Table,Furniture,4,7000
112,2026-03-19,Headphones,Electronics,3,2000
113,2026-04-02,Keyboard,Accessories,5,1000
114,2026-04-07,Mouse,Accessories,8,500
115,2026-04-12,Mobile,Electronics,4,20000
116,2026-04-17,Laptop,Electronics,2,50000
117,2026-05-03,Chair,Furniture,5,3000
118,2026-05-08,Table,Furniture,3,7000
119,2026-05-13,Headphones,Electronics,6,2000
120,2026-05-18,Keyboard,Accessories,7,1000
```

6. PYTHON IMPLEMENTATION

Procedure:

• Import Pandas and Matplotlib.

• Read the CSV file and inspect the data.

• Convert Order_Date to date format and calculate Sales.

• Group the data by product, month, and category.

• Create a bar chart, line chart, and pie chart.

• Observe the charts and record insights.

Program 1: Load and inspect the dataset

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("ecommerce_sales.csv")
df["Order_Date"] = pd.to_datetime(df["Order_Date"])
df["Sales"] = df["Quantity"] * df["Unit_Price"]

print(df.head())
print("Total Sales:", df["Sales"].sum())
print("Total Orders:", df["Order_ID"].nunique())
print("Total Products:", df["Product"].nunique())
```

Expected summary: Total Sales = ■580,000; Total Orders = 20; Unique Products = 7.

Program 2: Bar chart — total sales by product

```
product_sales = df.groupby("Product")["Sales"].sum().sort_values()

plt.figure(figsize=(8, 5))
product_sales.plot(kind="barh", color="steelblue")
plt.title("Total Sales by Product")
plt.xlabel("Total Sales (■)")
plt.ylabel("Product")
plt.tight_layout()
plt.show()
```

Output — Bar Chart

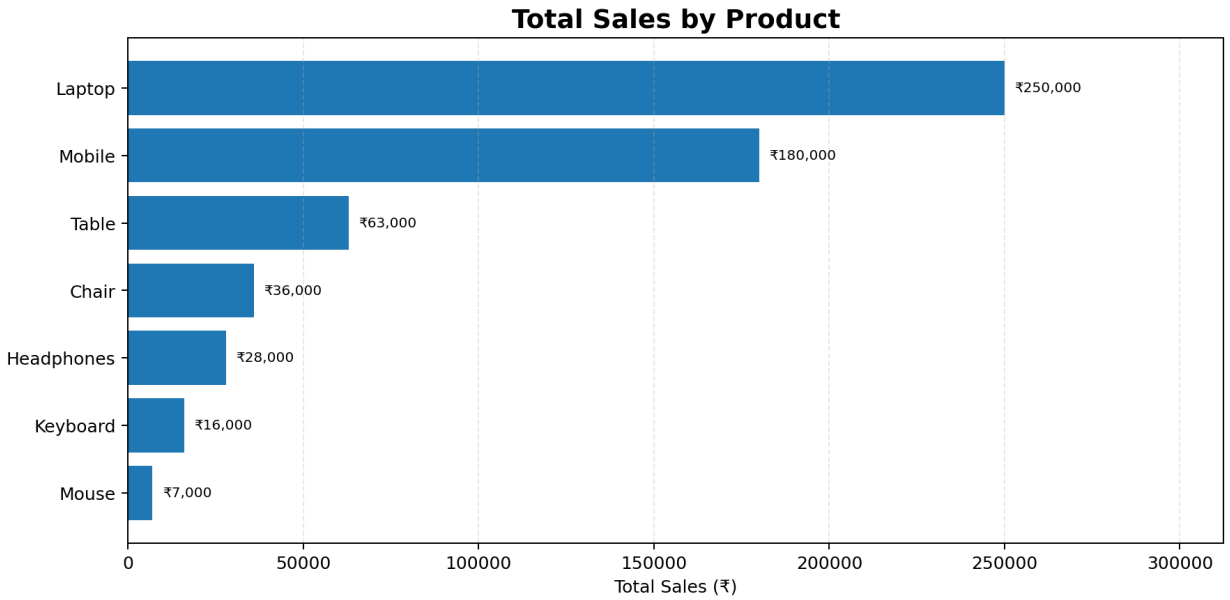


Figure 1: Product-wise sales. Laptop has the highest revenue in the sample dataset.

Program 3: Line chart — monthly sales trend

```
df["Month"] = df["Order_Date"].dt.to_period("M")
monthly_sales = df.groupby("Month")["Sales"].sum()

plt.figure(figsize=(8, 5))
plt.plot(monthly_sales.index.astype(str), monthly_sales.values,
         marker="o", color="darkorange")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales (■)")
plt.grid(True)
plt.tight_layout()
plt.show()
```

Output — Line Chart

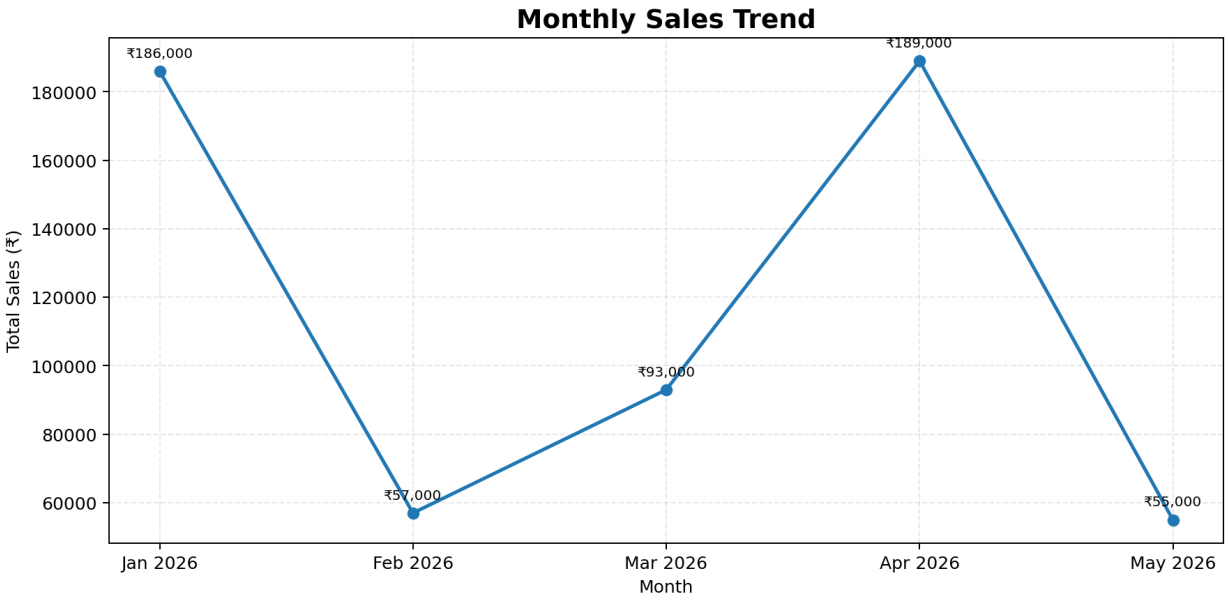


Figure 2: Monthly sales from January to May 2026.

Program 4: Pie chart — sales distribution by category

```
category_sales = df.groupby("Category")["Sales"].sum()

plt.figure(figsize=(7, 5))
plt.pie(category_sales.values, labels=category_sales.index,
        autopct="%1.1f%%", startangle=90)
plt.title("Sales Distribution by Category")
plt.tight_layout()
plt.show()
```

Output — Pie Chart

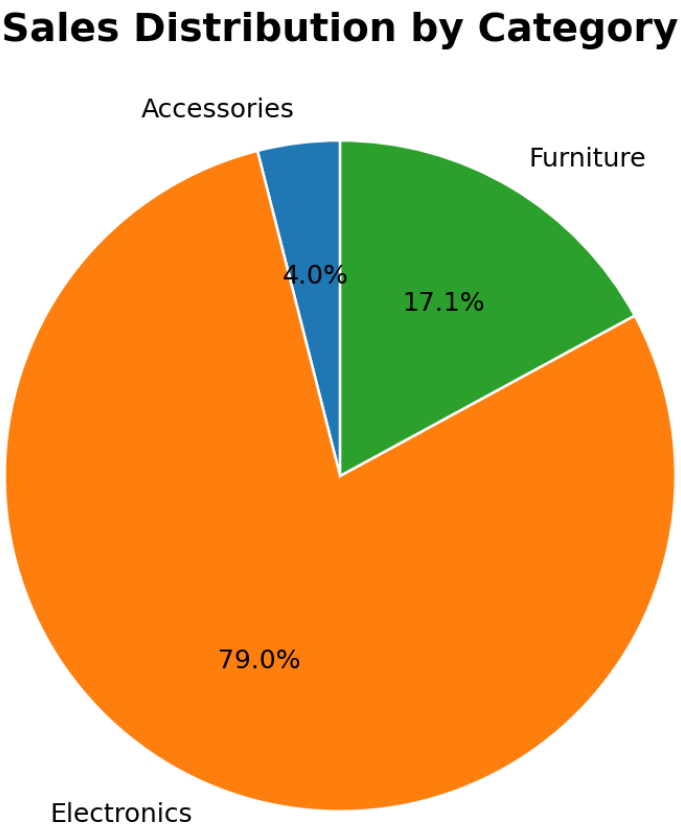


Figure 3: Category-wise contribution to total sales.

7. POWER BI IMPLEMENTATION

Procedure:

• Open Power BI Desktop and choose Home → Get Data → Text/CSV.

• Import ecommerce_sales.csv.

• Select Transform Data and check column data types.

• Set Order_Date to Date and Quantity and Unit_Price to Whole Number.

• Select Close & Apply.

• Choose Modeling → New column and create the Sales column.

• Create measures and add the charts to the report page.

Calculated column (DAX):

```
Sales = ecommerce_sales[Quantity] * ecommerce_sales[Unit_Price]
```

Measures (DAX):

```
Total Sales = SUM(ecommerce_sales[Sales])

Total Orders = DISTINCTCOUNT(ecommerce_sales[Order_ID])

Total Products = DISTINCTCOUNT(ecommerce_sales[Product])
```

Create the visualizations:

• Bar chart: Product → Y-axis; Sales → X-axis. Sort Sales descending.

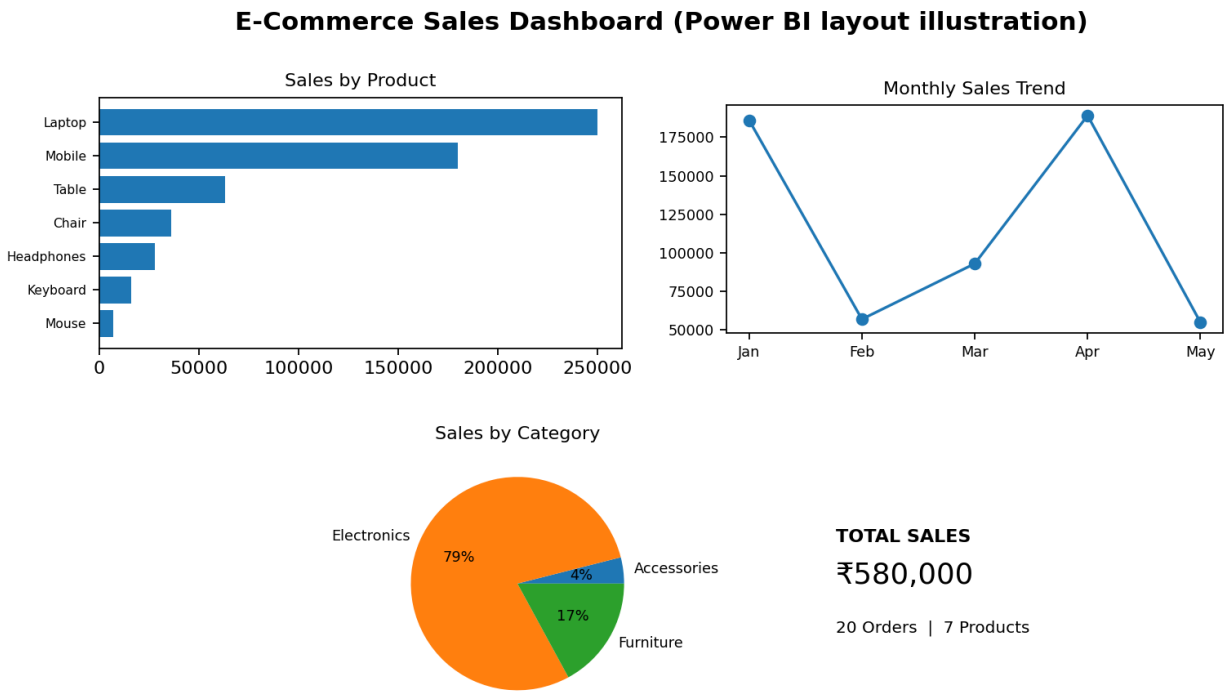
• Line chart: Order_Date (Month) → X-axis; Sales → Y-axis.

• Pie chart: Category → Legend; Sales → Values.

• Add Card visuals for Total Sales, Total Orders, and Total Products.

• Add titles, format visuals, and save as Ecommerce_Sales.pbix.

Dashboard Layout Illustration



Illustrative layout based on the sample data, not a screenshot from Power BI. Recreate these visuals in Power BI Desktop.

8. TABLEAU IMPLEMENTATION

Procedure:

• Open Tableau Desktop and select Connect → Text File.

• Open ecommerce_sales.csv.

• Create a calculated field named Sales using the formula below.

• Create three worksheets and combine them in a dashboard.

Calculated field:

```
[Quantity] * [Unit_Price]
```

Create the visualizations:

• Bar chart: Drag Product to Rows and Sales to Columns; use SUM(Sales).

• Line chart: Drag Order_Date to Columns and Sales to Rows; select Month and Line.

• Pie chart: Select Pie in Marks; drag Category to Color and Sales to Angle.

• Select New Dashboard and drag the three worksheets onto it.

• Add titles and save as Ecommerce_Sales.twbx.

Use the Python chart images as references while recreating the charts in Tableau.

9. VISUALIZATIONS

1. Bar Chart — Total Sales by Product
Compares revenue generated by each product and identifies the product with the highest sales.

2. Line Chart — Monthly Sales Trend
Shows how revenue changes from month to month and helps identify periods with higher or lower sales.

3. Pie Chart — Sales Distribution by Category
Shows the share of Electronics, Furniture, and Accessories in total revenue.

10. OBSERVATION AND INSIGHTS

Metric	Result
Total sales	■580,000
Total orders	20
Unique products	7
Highest-revenue product	Laptop — ■250,000
Lowest-revenue product	Mouse — ■7,000
Highest-revenue category	Electronics — ■458,000
Second-highest category	Furniture — ■99,000
Lowest-revenue category	Accessories — ■23,000

• Electronics contributes the largest share of revenue in the sample data.

• Laptop is the highest-revenue product in the sample dataset.

• Monthly sales vary based on the products and quantities ordered.

• Python, Power BI, and Tableau provide different ways to analyze and present the same sales data.

Note: All numerical insights are calculated from the sample dataset included in this report.

11. RESULT

The E-Commerce Sales dataset was analyzed using Python, Power BI, and Tableau. Bar, line, and pie charts were created to compare product sales, examine monthly revenue trends, and understand category-wise sales distribution. The project demonstrates how data visualization tools convert raw sales data into meaningful information that can support business decisions.